Balance disc design

COPY COMPLIANT to the original document resident on the computer system at the time of printing

Index

1. Purpose 3
2. Abbreviations. 3
3. Scope. 3
4. General procedure: description. 3
5. Calculation method. 5
5.1. Annular seal. 5
5.2. Restriction orifice. 7
5.3. Balance disc 8
5.4. Balance disc's axial thrust 9
6. Acceptance Criteria. 10
7. Design recommendations. 10
8. References 15
9. Appendix. 15
9.1. Calculation Tool. 15

COPY COMPLIANT to the original document resident on the computer system at the time of printing

1. Purpose

This document describes how to size a balance disc, mainly in terms of internal and external diameter, in horizontal multi-stage pump.

2. Abbreviations.

NDE Non Drive End

DE Drive End

D_2 Impeller external diameter

D_{bd} Balance disc external diameter

D_1 Balance disc internal diameter

Re Reynolds number

C_{ax} Fluid axial speed

3. Scope.

This procedure is applicable to horizontal multi-stage pumps equipped with balance disc.

4. General procedure: description.

The balance disc is a self-regulating device, which is able to completely balance the axial thrust generated by the pressure distribution on the impellers and shaft ends at all the operating flowrates.

For this reason, in case of significant axial thrust, it is generally used in clean services, such as boiler feed applications.

This device shall be used only in continuous service, since frequent start-up and shut-down could lead to impacts between disc and counterdisc and to their early wear and/or failure.

The considered acceptance criteria for the design is the one recommended by EPRI' Boiler Feedpump Operation and Maintenance Guidelines, according to which the balancing disc shall operate with a gap in the range of 50 µm and 100 µm (refer to section 7).

The calculations allow to define the geometry of the disc based on the operating conditions.

In general, the first preliminary sizing of the balancing disc diameters (both internal and external) is carried out based on internal practice.

COPY COMPLIANT to the original document resident on the computer system at the time of printing

$$D_{bd} \approx 0.75\, D_2$$

$$D_1 \approx 0.75\, D_{bd}$$

Where

- D_2 is the impeller external diameter;
- D_{bd} the disc external diameter;
- D_1 the disc internal diameter.

The input data required for the spreadsheet Mod.422 are the geometry and the operating conditions of the pump, the estimated axial thrust due the impellers and the pressure drop in the rear shroud of the last stage impeller.

The calculation is carried out with an iterative process, as per flow chart represented in

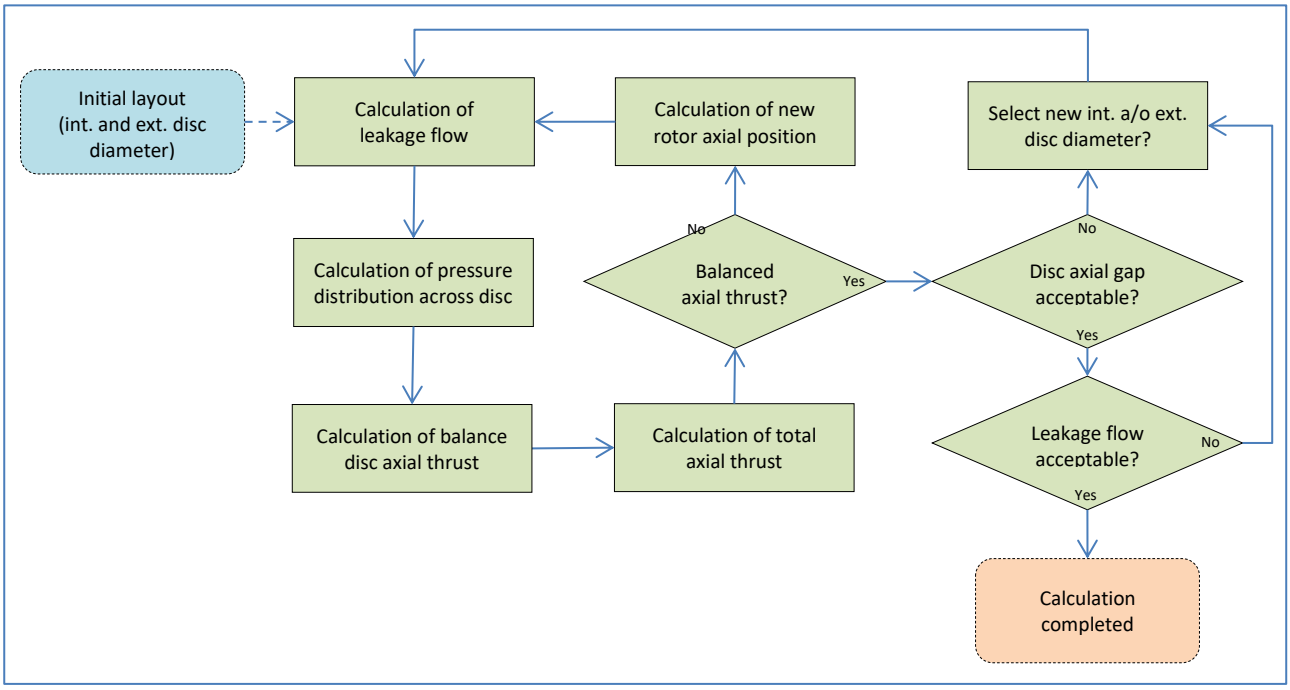


Figure 1 Flow chart.. The calculation is initialized with the following data:

1) Internal and external disc diameter: The recommended values are indicated above.
2) Disc axial gap: The algorithm automatically considers 75 μm as starting value, which is centered in EPRI's recommended range (refer to section 7).
3) Leakage flow: The algorithm automatically considers 1% of pump operating flowrate as starting value.

COPY COMPLIANT to the original document resident on the computer system at the time of printing

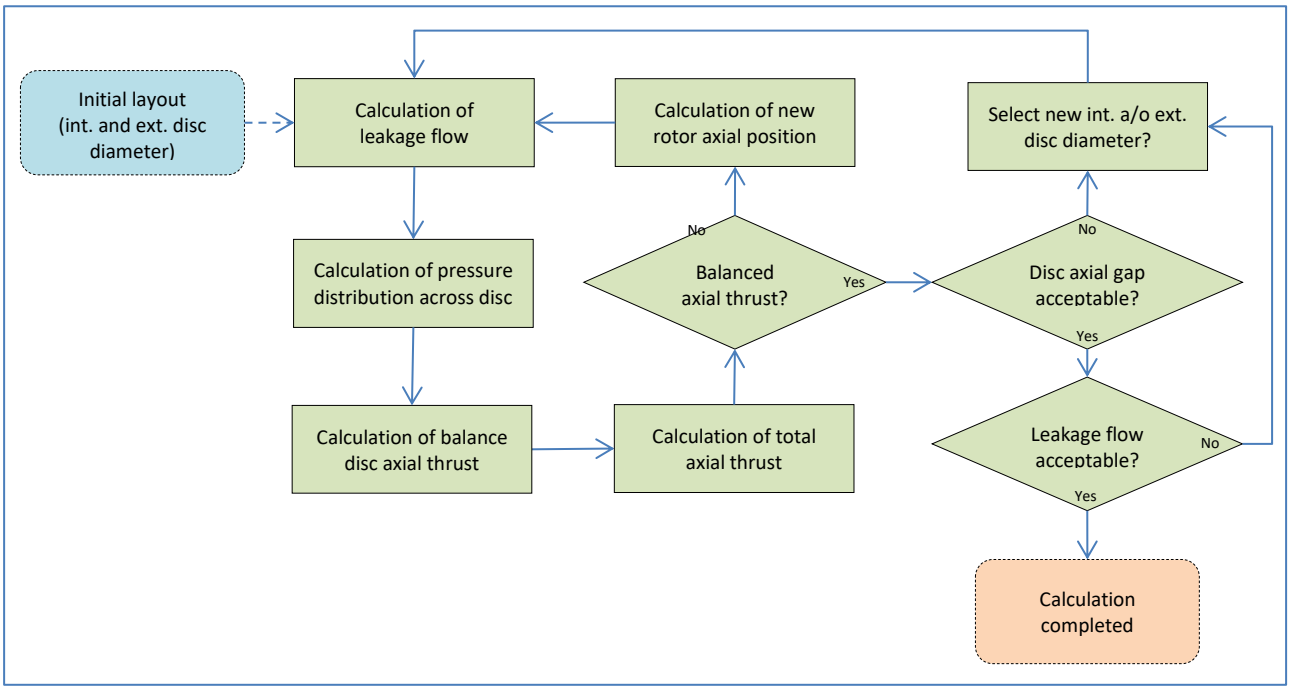


Figure 1 Flow chart.

5. Calculation method.

The hydraulic circuit considered in the calculation is composed by the following consecutive components: shaft bushing, balancing disc, balancing line with relevant restriction orifice, shaft seat on suction casing.

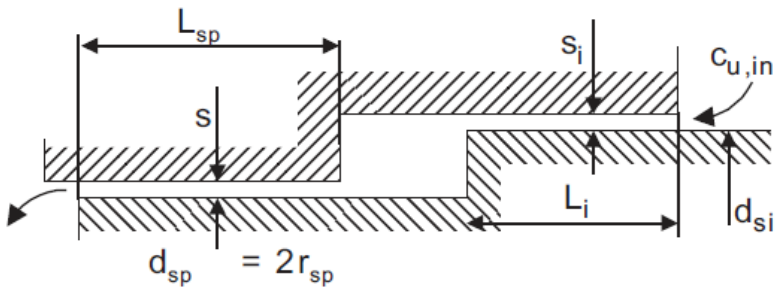
In the following the criteria adopted for the calculation are described.

5.1. Annular seal.

The leakages and the pressure drop across bushing and shaft seat on suction casing are studied on the basis of the “Centrifugal Pump” by J. F. Gulich.

The criteria are defined based on the following formulas (ref. J. F. Gulich’s Centrifugal Pumps 3.7.9).

COPY COMPLIANT to the original document resident on the computer system at the time of printing



$$Re_u = \frac{2\, s\, u_{sp}}{\nu}$$

$$u_{sp} = \frac{\pi\, d_{sp}\, n}{60}$$

$$Re = \frac{2\, s\, c_{ax}}{\nu}$$

Where:

- Re_u and Re are Reynolds numbers calculated based on tangential and axial fluid speeds respectively;
- s is the radial clearance of the annular seal;
- u_{sp} is the velocity of the external surface of the annular seal;
- d_{SP} the diameter of the annular seal;
- c_{ax} is the fluid axial speed flowing through the seal;
- v is the kinematic viscosity.

Based on the above-mentioned data, the friction coefficient is calculated (ref. J. F. Gulich's Centrifugal Pumps 3.7.11-12).

$$\lambda_0 = \frac{96}{Re}\left(1 - 0.6\frac{e_x}{s}\right), for\ laminar\ flow, use\ for\ Re < 2000\ (u_{sp} = 0)$$

$$\lambda_0 = \frac{0.31}{\left\{\log\left(A + \frac{6.5}{Re}\right)\right\}^2}, for\ turbulent\ flow, use\ for\ Re > 2000\ (u_{sp} = 0)$$

Seal surface	A
Rough	0.135 ε/s
Serrated	0.005 to 0.01

COPY COMPLIANT to the original document resident on the computer system at the time of printing

Isotropic pattern	0.01 to 0.03

Where λ_0 is the fiction coefficient for laminar (Re < 2000) or turbulent (Re > 2000) flow, not considering the rotational speed (u_{sp} = 0). In case of laminar – turbulent transition, it is recommended to use the maximum λ_0 among the above calculated ones. The term e_x is the eccentricity (usually equal to 0).

The effect of the rotational speed is evaluated in the definition of the corrected friction coefficient λ (ref. J. F. Gulich's Centrifugal Pumps 3.7.13-14).

$$\frac{\lambda}{\lambda_0} = 1 + 0.2\left(\frac{Re_u}{2000}\right)^{1.03}, influence\ of\ rotation\ for\ laminar\ flow\ (Re < 2000)$$

$$\frac{\lambda}{\lambda_0} = \left\{1 + 0.19\left(\frac{Re_u}{Re}\right)^2\right\}^{0.375}, influence\ of\ rotation\ for\ turbulent\ flow\ (Re > 2000)$$

The pressure drop across the annular seal is estimated through the following formula (ref. J. F. Gulich's Centrifugal Pumps 3.7.6).

$$c_{ax} = \sqrt{\frac{2\,\Delta p}{\zeta_{EA} + \lambda\frac{L_{sp}}{2s}}}$$

Where:

- c_{ax} is the fluid axial speed in the gap,
- L_{sp} the annular seal's length,
- s the radial gap
- ζ_{EA} the inlet and outlet losses coefficient (from 1 to 1.2).

5.2. Restriction orifice.

The leakage flow and the pressure drop across the restriction orifice are estimated based on the ISO 5167-2 Standard.

The discharge coefficient C is given by Reader-Harris/Gallagher's equation (ref. ISO 5167-2 sect. 5.3.2.1).

COPY COMPLIANT to the original document resident on the computer system at the time of printing

$$C = 0.5961 + 0.0261\beta^2 - 0.216\beta^8 + 0.000521\left(\frac{10^6\beta}{Re_D}\right)^{0.7} + (0.0188 + 0.0063A)\beta^{3.5}\left(\frac{10^6}{Re_D}\right)^{0.3} +$$

$$+(0.043 + 0.80e^{-10L_1} - 0.123e^{-7L_1})(1 - 0.11A)\frac{\beta^4}{1-\beta^4} - 0.031\left(M_2 - 0.8{M_2}^{1.1}\right)\beta^{1.3}$$

If D < 71.12mm (2.8 inch), the following term shall be added to the above reported equation:

$$+0.011(0.075 - \beta)\left(2.8 - \frac{D}{25.4}\right)$$

Where:

- β is the ratio between the orifice diameter d and the pipe diameter D;
- Re_D is the Reynolds number calculated on the pipe section;
- $L_1 = l_1/D$ with l_1 the distance of the upstream tapping from the upstream face of the plate;
- $L_2 = l_2/D$ with l_2 the distance of the downstream tapping from the downstream face of the plate;
- M_2 and A are defined as in the following:

$$M_2 = \frac{2L_2}{1-\beta}$$

$$A = \left(\frac{19000\beta}{Re_D}\right)^{0.6}$$

In the application, the coefficients L_1, L_2 are usually equal to zero.

The pressure drop across the orifice is extracted from the following (ref. ISO 5167-2 sect. 4).

$$q_m = \frac{C}{\sqrt{1-\beta^4}}\varepsilon\frac{\pi}{4}d^2\sqrt{2\Delta p\,\rho_1}$$

Where q_m is the mass flowrate; ε is the expansibility factor, whose effect is negligible (equal to 1) for the applicable cases of this procedure.

5.3. Balance disc

The balancing disc is studied based on J. F. Gulich's formulas for the calculation of leakages in the radial gap (ref. J. F. Gulich's Centrifugal Pumps 3.7.21).

$$Re = \frac{2\,s\,c_i}{\nu}\sqrt{a_r}$$

COPY COMPLIANT to the original document resident on the computer system at the time of printing

$$Re_u = \frac{2\, s\, \omega\, r_a}{\nu}\sqrt{a_r}$$

$$a_r = \frac{r_i}{r_a}$$

Where:

- Re_u and Re are Reynolds numbers calculated on the basis of tangential and radial fluid speeds respectively;
- c_i is the fluid radial speed, r_r and r_a are the respectively the internal and the external dis diameters;
- s is the axial gap between disc and counter disc;
- ω is the rotational speed;
- ν is the fluid kinematic viscosity.

Based on the above-mentioned data, the friction coefficient is calculated (ref. J. F. Gulich's Centrifugal Pumps 3.7.11-12).

$$\lambda_0 = \frac{96}{Re}\left(1 - 0.6\frac{e_x}{s}\right), for\ laminar\ flow, use\ for\ Re < 2000\ (u_{sp} = 0)$$

$$\lambda_0 = \frac{0.31}{\left\{\log\left(A + \frac{6.5}{Re}\right)\right\}^2}, for\ turbulent\ flow, use\ for\ Re > 2000\ (u_{sp} = 0)$$

Seal surface	A
Rough	0.135 ε/s
Serrated	0.005 to 0.01
Isotropic pattern	0.01 to 0.03

Where λ_0 is the fiction coefficient for laminar (Re < 2000) or turbulent (Re > 2000) flow, not considering the rotational speed (u_{sp} = 0). In case of laminar – turbulent transition, it is recommended to use the maximum λ_0 among the above calculated ones.

The effect of the rotational speed is evaluated in the definition of the corrected friction coefficient λ (ref. J. F. Gulich's Centrifugal Pumps 3.7.13-14).

COPY COMPLIANT to the original document resident on the computer system at the time of printing

$$\frac{\lambda}{\lambda_0} = 1 + 0.2\left(\frac{Re_u}{2000}\right)^{1.03}, influence\ of\ rotation\ for\ laminar\ flow\ (Re < 2000)$$

$$\frac{\lambda}{\lambda_0} = \left\{1 + 0.19\left(\frac{Re_u}{Re}\right)^2\right\}^{0.375}, influence\ of\ rotation\ for\ turbulent\ flow\ (Re > 2000)$$

The pressure drop across the annular seal is estimated through the following formula (ref. J. F. Gulich's Centrifugal Pumps 3.7.6).

$$c_{ax} = \sqrt{\frac{2\,\Delta p}{\zeta_{EA} + \lambda\frac{L_{sp}}{2s}}}$$

Where:

- c_{ax} is the fluid axial speed in the gap;
- L_{sp} is the annular seal's length;
- S is the radial gap;
- ζ_{EA} is the inlet and outlet losses coefficient (from 1 to 1.2).

5.4. Balance disc's axial thrust

The axial thrust acting on the balance disc is calculated referring to internal practice according to the following formula:

$$F_{bd} = \frac{\pi}{3}\Delta p_2\left(\frac{D_{bd}}{2}\right)^2 [1 + \rho_1 + {\rho_1}^2 - 3{\rho_b}^2 - \alpha(1 + \rho_1 - 2{\rho_1}^2)]$$

Where:

- Δp_2 is the difference between pressure upstream the disc and the intermediate pressure downstream the disc;
- ρ_1 is the ratio between internal and external diameter of balancing disc;
- ρ_b is the ratio between the piston (installed upstream the disc) and external disc diameter;
- D_{bd} is the external disc diameter.

The calculated axial thrust shall be equal in intensity and opposite in direction respect to the axial thrust generated by the rotor.

6. Acceptance Criteria.

COPY COMPLIANT to the original document resident on the computer system at the time of printing

It is recommended that the axial gap between disc and counter disc is included between the recommended values as per section 4.

If the calculated disc diameter significantly differs from the recommended value from internal practice, a deeper investigation is required (contact Engineering dpt. or R&D dept.).

Based on the pump performances, the leakage flow should be about 1% of the pump operating flowrate.

Refer to the flow chart at

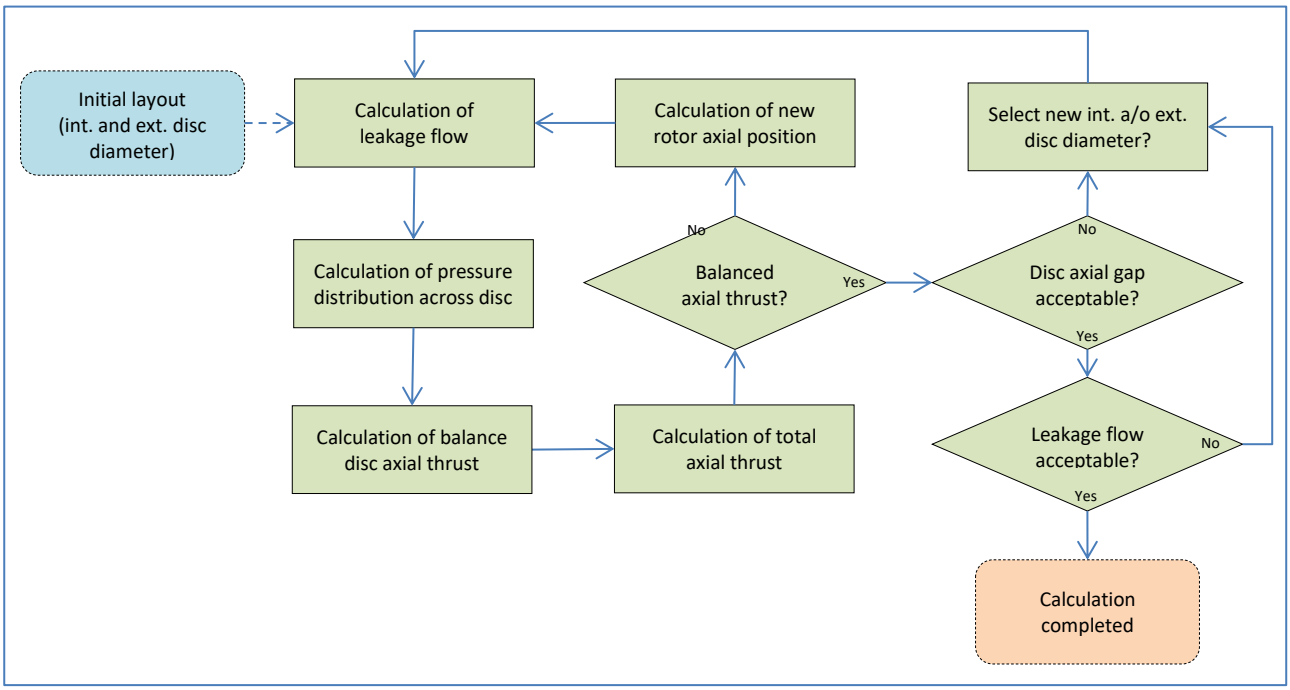


Figure 1 Flow chart.for more details.

7. Design recommendations.

The primary mechanism of balancing disc failure is represented by the contact between the rotating disc and the relevant counter disc. For this reason, minimum axial gap shall be defined and considered during the design of the balancing disc system.

In addition, excessive axial gap highly affects the leak-off flow and the volumetric efficiency of the pump, which requires to define an upper limit during the design.

According to Boiler Feedpump Operation and Maintenance Guidelines by EPRI, the balancing disc shall operate with a gap in the range of 50 µm and 100 µm.

COPY COMPLIANT to the original document resident on the computer system at the time of printing

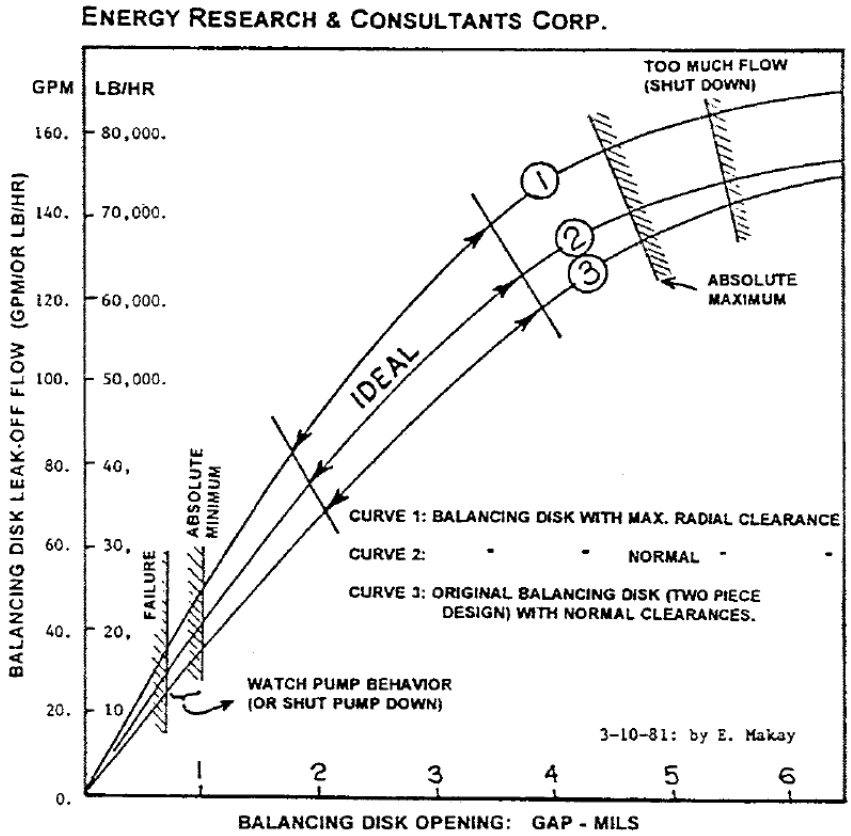


Figure 2 Guidelines for balancing disc leak-off flows for safe operation of boiler feed pumps (EPRI).

The lower and upper limits are significantly related to the dynamic stiffness of the balancing disc also. As indicated in Figure 3 Dynamic stiffness of balance disc., the dynamic stiffness for plain disc (parallel face) could be negative for very low axial gap. In case of an external perturbation, the system would be unstable and the gap could reduce to zero with the relevant contact between disc and counter disc. For this reason, a lower limit for the axial gap shall be defined to avoid this failure.

In addition, for greater gap, the dynamic stiffness is very low (close to zero). This case is not recommended, since external perturbation could lead to significant gap's variation, which requires the definition of a gap's upper limit.

COPY COMPLIANT to the original document resident on the computer system at the time of printing

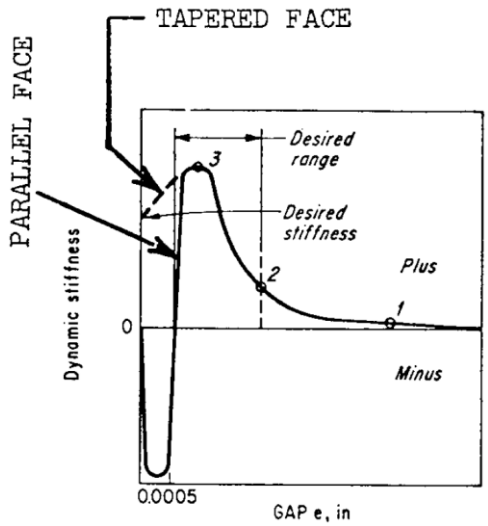


Figure 3 Dynamic stiffness of balance disc.

A way to reduce this unstable effect is to design the disc with tapered face, as per Figure 4 Balance disc with parallel faces. Force equilibrium is shown when disc in normal open position. and Figure 5 Tapered face balance disc.. In this case, the dynamic stiffness is positive in all the operating conditions, resulting in a more stable solution.

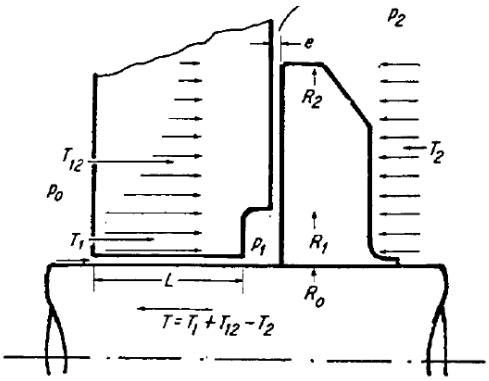


Figure 4 Balance disc with parallel faces. Force equilibrium is shown when disc in normal open position.

COPY COMPLIANT to the original document resident on the computer system at the time of printing

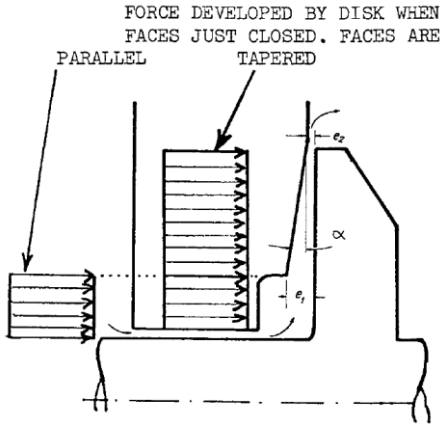


Figure 5 Tapered face balance disc.

As per Figure 6 Balancing disc thrust carrying capability and leak-off flow., the tapered solution is able to carry a greater thrust for low gap values.

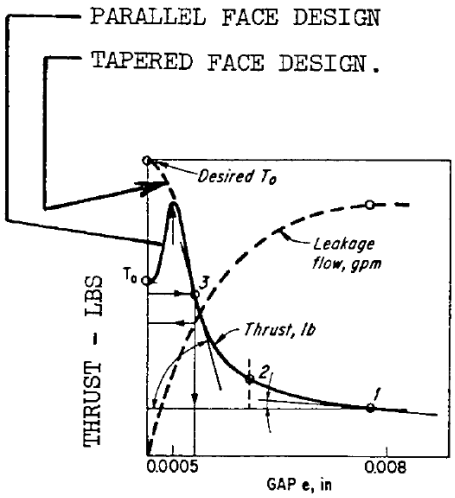


Figure 6 Balancing disc thrust carrying capability and leak-off flow.

The stability of the balancing disc is affected by the chamber upstream the disc. The tested data show that great chamber in terms could lead to vibration and instability of the disc, due to the fluid pressure fluctuations downstream the last stage impeller. For this reason, it is recommended to reduce the size of this chamber (refer to the following Figure 7 Close chamber.and Figure 8 Open chamber.).

COPY COMPLIANT to the original document resident on the computer system at the time of printing

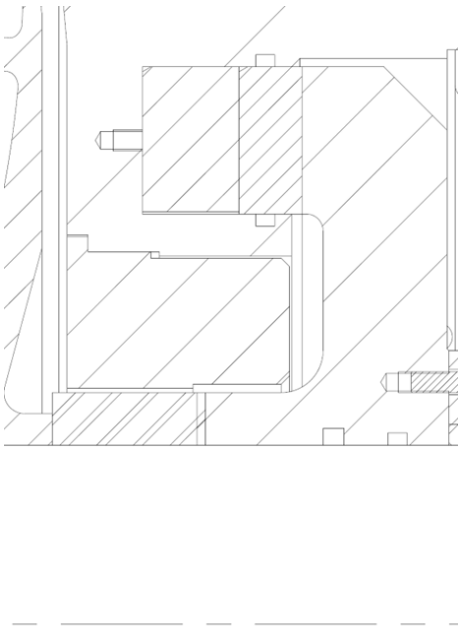


Figure 7 Close chamber.

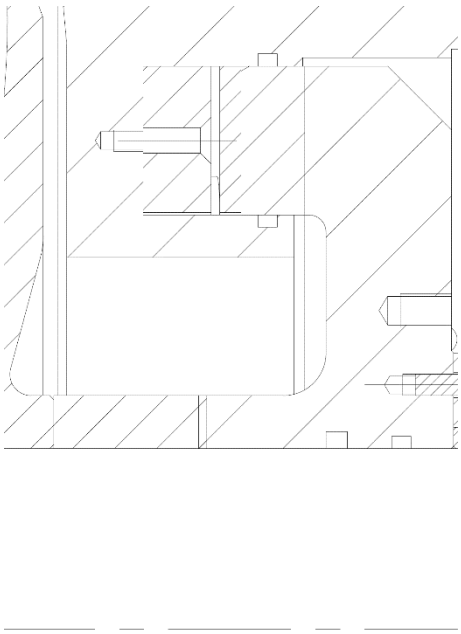


Figure 8 Open chamber.

In addition to the above, some design recommendations apply to the counter disc.

A dedicated O-ring seal shall be foreseen between counter disc and pump casing, in order to avoid possible leakages in the clearance, which could compromise its securing and fixation. In

COPY COMPLIANT to the original document resident on the computer system at the time of printing

general, dedicated securing screws (centered in the surface of the counter disc) are usually recommended.

The material used for the balance disc and relevant counterdisc shall be selected with the same approach as per the other wear parts. The material shall have good wear resistance and anti-galling properties. For this reason, they are usually selected among the following options:

- Steel materials with hardening a/o hard facing, on the basis of the feasibility;
- Other wear resistance composite material reinforced with carbon fiber (such as PEEK). In this case, it recommended to foresee a proper gap between counterdisc and steel casing at the external diameter, in order to have proper flushing and to avoid galling phenomena due to counterdisc deformation during operation.

As previously indicated, the balance disc is not recommended in case of intermittent service, since contacts a/o impacts could occur between disc and counterdisc, leading to their early wear and/or failure. For this reason, it is recommended to provide a suitable lift-off device. This device keeps the balance disc open during the stand-by, start-up and shut-down phases of the pump, by means of preloaded springs acting on the rotor. It provides high reliability for pumps, intended to be operated with frequent starts and stops.

8. References

Centrifugal Pumps – J. F. Gulich

Hydraulic axial thrust balancing – balancing disc – Turboinstitut

ISO 5167-2

Boiler Feedpump Operation and Maintenance Guidelines – EPRI

Module Mod.422

9. Appendix.

9.1. Calculation Tool.

The following instructions shall be considered to carry out the calculation by means of the dedicated tool Mod.422:

1) Compile all the input data in the following sections to allow the initialization of the calculation:
    a. Operating data.
    b. Geometric data of bushing between last stage impeller and disc.

COPY COMPLIANT to the original document resident on the computer system at the time of printing

c. Geometric data of balancing disc.

d. Geometric data of restriction orifice installed in the balancing line.

e. Geometric data of casing seat for the shaft.

As general note, the loss coefficients in all the sections have already been pre-compiled. It is not recommended to modify them, unless it is necessary for sensitivity analysis.

The calculation is automatically initialized with disc axial gap 75 µm and leakage flow 1% of pump operating flowrate.

2) Run the calculation pushing the dedicated button. The tool consider 0.1% accuracy and 1000 iterations as default initial parameter. If the results are not acceptable (refer to following steps), these values can be modified.

CALCULATION PARAMETERS		
iterations	1000	Number of max iterations for the calculation
accuracy [%]	0.100	Required accuracy
Solution is updated		

Figure 9 Calculation parameters.

3) Check the results in the "Calculated data" section. The results are acceptable on the basis of criteria at Figure 11 Acceptance criteria..

CALCULATED DATA		
$Q_{leakage}$ [m³/h]	3.483	Leakage flow
$Q_{leakage}$ [%]	0.013	Volmetric loss
$P_{bushing}$ [barg]	32.240	Pressure downstream bushing
P_{disc} [barg]	10.10	Pressure downstream disc
P_{disc}* [barg]	8.376	Pressure downstream disc at shaft diameter
$P_{stuffing\ box}$ [barg]	1.77	Stuffing box pressure
$P_{suction}$* [barg]	1.500	Calculated suction pressure
$\Delta P_{verification}$ [%]	0.009	Check for the calculation
ΔF_{tot} [N]	-38	Residual axial thrust
ΔF_{tot} [%]	-0.084	Total axial thrust
r_a* [mm]	101.25	Reference external disc radius as per internal practice
r_a/r_a* [%]	13.6%	Difference between selected and Turboinstitut external disc radii
r_i* [mm]	75.94	Reference disc's internal radius as per Turboinstitut
r_i/r_i* [%]	5.3%	Difference between selected and Turboinstitut internal disc radii
s [mm]	0.0982	Axial gap between disc and couterdisc
s/s_{min} [%]	196%	Ratio between calculated and min recommended axial gap (0.05 mm)
s/s_{max} [%]	98%	Ratio between calculated and min recommended axial gap (0.1 mm)

Figure 10 Results summary.

COPY COMPLIANT to the original document resident on the computer system at the time of printing

Warnings legend	
	= Acceptable data.
	= Not optimal data. User shall modify input data to have fully acceptable solution (green). If not possibile, yellow fields can be accepted as well.
	= Not acceptable data. User shall modify input data to find an acceptable solution.

Figure 11 Acceptance criteria.

4) If all the results are acceptable, the calculation is completed. Otherwise, input diameters of the disc shall be modified and the calculation shall be run again.
5) Accuracy can be updated, if necessary. If the result is not aligned with the set accuracy, the number of iterations shall be increased.

COPY COMPLIANT to the original document resident on the computer system at the time of printing